

From unit-cell physics to design choices

Why is maximum lattice multiplication not the same as the best reactor design?

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LESSON

Read: Continue § 4.5: Eqs. (4.29)–(4.40), (4.48)–(4.55), Table 4.3, and the PWR example/Fig. 4.7.

1 Connect lump size to resonance escape

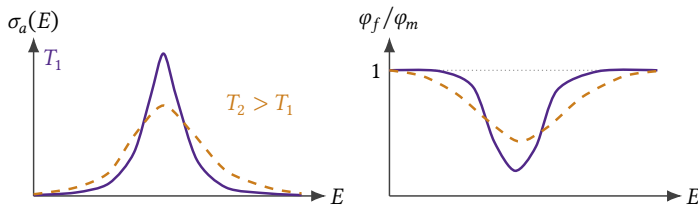
The heterogeneous resonance integral is

$$I = \int_I \sigma_a^{fe}(E) \frac{\varphi_f(E)}{\varphi_m(E)} \frac{dE}{E}.$$

Slowing down mainly in the moderator, an approximately $1/E$ moderator spectrum between resonances, and small losses at each resonance lead to

$$p \approx \exp \left[-\frac{V_f N_{fe} I}{V_m \xi_m \Sigma_s^m} \right].$$

The exponential combines successive resonance survival probabilities. For a mixture, use $\xi \Sigma_s = \sum_i \xi_i \Sigma_{s,i}$. Spatial self-shielding lowers φ_f/φ_m at resonance energies, reducing I . This is the quantitative link back to Lesson 15's uranium lump.



Adapted from Fig. 4.6: cooler fuel (solid), hotter fuel (dashed). Schematic only; the relevant loss is the integral of $\sigma_a \varphi_f$, not the peak cross section.

Doppler broadening spreads absorption into less-shielded wings. Thus, raising fuel temperature increases effective resonance absorption and lowers p .

Empirical Resonance Integral for UO_2 Fuel Elements

Table 4.3 gives, for isolated UO_2 rods,

$$I [\text{b}] = 4.45 + 26.6 \sqrt{\frac{4}{\rho_f D}}.$$

Insert ρ_f in g/cm^3 and D in cm; the stated range is $0.2 < D < 3.5$ cm.

2 Translate PWR pitch into atom ratios

For the square UO_2 -cladding-water cell of Lesson 17, with gap neglected and $D = 2r_f$,

$$\frac{V_m}{V_f} = \frac{P^2 - \pi r_c^2}{\pi r_f^2},$$

$$\frac{H}{U} = \frac{N_H^m}{N_U^f} \frac{P^2 - \pi r_c^2}{\pi r_f^2}, \quad \frac{P}{D} > \frac{r_c}{r_f}.$$

At fixed radii and material densities, increasing P/D increases the moderator-to-fuel ratio. Note that changing D instead also changes self-shielding and heat conduction.

3 Find the competing effects

More moderator: resonance escape p generally rises, but thermal utilization f falls as more neutrons are absorbed outside fuel. Their competition produces the maximum in k_∞ discussed with Fig. 4.7.

More enrichment: η_T rises as a larger share of fuel absorptions occurs in fissile nuclei. For fissile/fertile fuel with negligible fertile thermal fission,

$$\eta_T \approx \frac{\tilde{e} \bar{v} \sigma_{fT}^{fi}}{\tilde{e} \bar{\sigma}_{aT}^{fi} + (1 - \tilde{e}) \bar{\sigma}_{aT}^{fe}}.$$

More soluble boron: greater nonfuel thermal absorption lowers f and changes the optimum. Fuel size, spectrum, and temperature can also alter the other factors; do not treat them as fixed constants in a transport sweep.

4 Separate an optimum from a design

An **undermoderated** cell lies on the low-moderator side of the maximum. There, decreasing moderator density tends to lower k_∞ . This is the desired moderator-density contribution to negative feedback, not a guarantee for the full temperature coefficient. Boron dilution by expansion, fuel Doppler broadening, and other effects must be separated.

Compare the two design levers. For an SFR, use enrichment to meet an assigned k_∞ target (Lesson 16). For a PWR, map the P/D optimum at fixed fuel composition. Both choices remain constrained by heat removal, mechanical spacing, control, leakage, and fuel depletion.

OpenMC | One controlled change per run

For the PWR, hold fuel/clad radii, enrichment, temperatures, and water density fixed while scanning P ; record $k_\infty \pm \sigma_k$, the four factors, and ζ . Repeat with boron, then vary fuel temperature and moderator density *separately*. Use H-in- H_2O thermal scattering data. Compare with the SFR enrichment sweep. **Explain:** what improves neutron economy, what changes feedback, and what can't be modeled by the unit cell?